

Theory of Computation

Lecture 6: Non-Regular Languages

The Pumping Lemma

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Finite automata are powerful.

But finite memory has fundamental mathematical limits.

Today's Roadmap

1. **The Graph Intuition:** Why finite states force repetition.
2. **The Formal Theorem:** Sipser's Pumping Lemma.
3. **The Adversary Game:** How to navigate the nested quantifiers.
4. **The Proof Template:** A rigorous, step-by-step recipe for contradictions.
5. **Worked Examples:** Proving languages are non-regular.

A Suspicious Language

Consider the language of perfectly balanced binary strings:

$$B = \{0^n 1^n \mid n \geq 0\}$$

To accept 00001111, a machine must somehow “remember” that it saw exactly four 0s so it can verify the four 1s.

The problem: A DFA only has a fixed number of states (say, p states). What happens when we feed it $p + 1$ zeros?

The Pigeonhole Principle in Directed Graphs

A DFA is simply a directed graph. Reading a string is just tracing a walk through that graph.

Pigeonhole Principle: If you put $m + 1$ pigeons into m holes, at least one hole contains two pigeons.

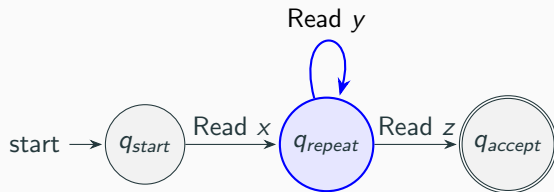
If a DFA has p states, and it reads a string of length p or greater, it visits at least $p + 1$ states (counting the start state). Therefore, **it must visit at least one state twice.**

Revisiting a State Creates a Cycle

In graph theory, a walk that intersects itself contains a cycle.

Let the string be $s = xyz$.

- x : The path from the start state to the first time we hit the repeated state.
- y : The cycle (leaving the state and returning to it).
- z : The path from the repeated state to the accept state.



The “Pump”

Because y is a cycle, the DFA doesn't actually know how many times it loops. If the DFA accepts xzy , it has no choice but to also accept:

- xz (looping 0 times, skipping y)
- $xyyz$ (looping 2 times)
- $xyyyz$ (looping 3 times)

Core Intuition: Any sufficiently long string accepted by a regular language **MUST** contain a cycle that can be “pumped” (repeated) infinitely without leaving the language.

The Pumping Lemma for Regular Languages

If A is a regular language, then there exists a number p (the pumping length) such that every string $s \in A$ with $|s| \geq p$ can be divided into three pieces, $s = xyz$, satisfying the following three conditions:

1. for each $i \geq 0$, $xy^iz \in A$,
2. $|y| > 0$, and
3. $|xy| \leq p$.

Why $|y| > 0$? The cycle must consume at least one character, or we aren't pumping anything.

Why $|xy| \leq p$? By the Pigeonhole Principle, the first duplicate state **MUST** occur within the first p steps.

The Logic of Contradiction

The Pumping Lemma is an implication:

IF A is regular $\implies A$ has the pumping property.

By modus tollens, if A does **not** have the pumping property, it is **not** regular.

The lemma relies on heavily nested quantifiers:

$$\exists p, \forall s, \exists x, y, z, \forall i \dots$$

To prove a language is NOT regular, we must negate this statement:

$$\forall p, \exists s, \forall x, y, z, \exists i \dots$$

How to play The Adversary Game

Think of the proof as a turn-based game against the SETUP DEMON who claims A is regular. You win if you force a contradiction.

Turn	Action
1. Demon:	Chooses the pumping length p . (You don't know p , it's a variable).
2. YOU:	Choose a target string $s \in A$ such that $ s \geq p$.
3. Demon:	Divides your string into $s = xyz$ however they want, as long as $ y > 0$ and $ xy \leq p$.
4. YOU:	Choose a pump count $i \geq 0$ such that $xy^iz \notin A$.

Rule of Rigor: You cannot choose the split (x, y, z) . You must prove your i works for *any* valid split the Demon could possibly make.

The Rigorous 5-Step Proof Template

Every valid Pumping Lemma proof follows exactly this structure. **Do not skip steps.**

1. **Assume for contradiction** that language L is regular.
2. Let p be the pumping length given by the Pumping Lemma.
3. **Choose a specific string** $s \in L$ where $|s| \geq p$.
(*This is your creative step. Express s in terms of p .*)
4. State that by the lemma, s can be split into $s = xyz$ such that $|y| > 0$ and $|xy| \leq p$.
Analyze what y must look like based on the $|xy| \leq p$ constraint.
5. **Pump** the string. Choose an $i \geq 0$. Show mathematically that $xy^iz \notin L$, creating a contradiction. Conclude L is not regular.

Example 1: Matching Two Blocks

$$L_1 = \{0^n 1^n \mid n \geq 0\}.$$

1. **Assume for contradiction** that L_1 is regular.
2. Let p be its pumping length.
3. Choose

$$s = 0^p 1^p.$$

Clearly, $s \in L_1$ and $|s| = 2p \geq p$.

4. For every valid split $s = xyz$ with $|xy| \leq p$ and $|y| > 0$, both x and y lie inside the first block of 0s. Hence

$$y = 0^k \quad \text{for some } 1 \leq k \leq p.$$

Example 1: Pumping Breaks Equality

5. Choose $i = 0$. Then

$$xy^0z = xz = 0^{p-k}1^p.$$

The pumped string contains $p - k$ zeros but still contains p ones.

Since $k > 0$,

$$p - k \neq p.$$

Therefore,

$$xz \notin L_1.$$

This contradicts the Pumping Lemma. Therefore, L_1 is not regular. ■

Intuition

The machine is forced to repeat or remove some early 0s without changing the later 1s.

Example 2: Palindromes

$$L_2 = \{w \in \{a, b\}^* \mid w = w^R\}.$$

1. **Assume for contradiction** that L_2 is regular.
2. Let p be its pumping length.
3. Choose

$$s = a^p b a^p.$$

This string is a palindrome and $|s| = 2p + 1 \geq p$.

4. For every valid split $s = xyz$, the condition $|xy| \leq p$ forces y to lie entirely inside the first block of as . Thus,

$$y = a^k \quad \text{for some } 1 \leq k \leq p.$$

Example 2: Pumping Breaks Symmetry

5. Choose $i = 0$. Then

$$xy^0z = a^{p-k}ba^p.$$

The left side of the central b has $p - k$ copies of a , while the right side has p copies.

Because $k > 0$, the two sides are unequal. Hence the string is not a palindrome:

$$xy^0z \notin L_2.$$

This contradicts the Pumping Lemma. Therefore, L_2 is not regular. ■

Intuition

Pumping changes only the left half, so the mirror image no longer matches.

Example 3: Synchronizing Three Blocks

$$L_3 = \{a^n b^n c^n \mid n \geq 0\}.$$

1. **Assume for contradiction** that L_3 is regular.
2. Let p be its pumping length.
3. Choose

$$s = a^p b^p c^p.$$

Clearly, $s \in L_3$ and $|s| = 3p \geq p$.

4. For every valid split $s = xyz$, $|xy| \leq p$ places y completely inside the initial block of as . Therefore,

$$y = a^k \quad \text{for some } 1 \leq k \leq p.$$

Example 3: One Block Changes, Two Do Not

5. Choose $i = 2$. Then

$$xy^2z = a^{p+k}b^pc^p.$$

The numbers of as , bs , and cs are now

$$p+k, \quad p, \quad p.$$

Since $k > 0$, these three quantities are not equal. Thus,

$$xy^2z \notin L_3.$$

This contradicts the Pumping Lemma. Therefore, L_3 is not regular. ■

Intuition

A finite automaton cannot keep three arbitrarily large blocks perfectly synchronized.

Example 4: Copying Across a Separator

$$\Sigma = \{0, 1, \#\} \quad L_4 = \{w\#w \mid w \in \{0, 1\}^*\}.$$

1. **Assume for contradiction** that L_4 is regular.
2. Let p be its pumping length.
3. Choose

$$s = 0^p 1 \# 0^p 1.$$

Here the same word $0^p 1$ appears on both sides of $\#$, so $s \in L_4$. Also, $|s| = 2p + 3 \geq p$.

4. For every valid split $s = xyz$, $|xy| \leq p$ forces y into the first block of 0s. Hence

$$y = 0^k \quad \text{for some } 1 \leq k \leq p.$$

Example 4: Pumping Breaks the Copy

5. Choose $i = 0$. Then

$$xy^0z = 0^{p-k}1\#0^p1.$$

The word to the left of $\#$ is now $0^{p-k}1$, while the word to the right is still 0^p1 .

Since $k > 0$, the two words are different. Therefore,

$$xy^0z \notin L_4.$$

This contradicts the Pumping Lemma. Therefore, L_4 is not regular. ■

Intuition

Pumping edits the first copy but leaves the second copy untouched.

Example 5: Unary Strings of Square Length

$$L_5 = \{a^{n^2} \mid n \geq 0\}.$$

1. **Assume for contradiction** that L_5 is regular.
2. Let $p \geq 1$ be its pumping length.
3. Choose

$$s = a^{p^2}.$$

Since p^2 is a perfect square, $s \in L_5$, and $|s| = p^2 \geq p$.

4. Every valid split $s = xyz$ has

$$y = a^k \quad \text{for some } 1 \leq k \leq p.$$

Example 5: Pumping Lands Between Two Squares

5. Choose $i = 2$. The pumped string has length

$$|xy^2z| = p^2 + k.$$

Since $1 \leq k \leq p$,

$$p^2 < p^2 + k \leq p^2 + p < (p+1)^2.$$

Thus, $p^2 + k$ lies strictly between the consecutive squares p^2 and $(p+1)^2$.

Therefore, $p^2 + k$ is not a perfect square, so

$$xy^2z \notin L_5.$$

This contradicts the Pumping Lemma. Therefore, L_5 is not regular. ■

Intuition

Even over a one-symbol alphabet, a DFA cannot recognize widely growing gaps between valid lengths.

What the Five Examples Teach Us

Language type	What must be remembered	What pumping breaks
$0^n 1^n$	one count	equality of two blocks
Palindromes	distant symbols	left-right symmetry
$a^n b^n c^n$	three counts	three-way synchronization
$w \# w$	an entire word	exact copying
a^{n^2}	special lengths	arithmetic spacing

Although the languages look different, every proof uses the same five moves:

assume $\rightarrow p \rightarrow s \rightarrow \text{trap } y \rightarrow \text{pump}.$

Checklist for Student Proofs

When grading your own proofs, ask:

1. Did I say “let p be the pumping length”? (Do not invent a value such as $p = 5$.)
2. Did I choose a string s using p ?
3. Did I verify both $s \in L$ and $|s| \geq p$?
4. Did I use $|xy| \leq p$ to prove where y must occur?
5. Did I handle **every** valid split rather than choosing y myself?
6. Did I select an i and clearly prove that $xy^iz \notin L$?

Closure Proof 1: When Pumping Becomes Tedious

Consider

$$L_{\neq} = \{0^m 1^n \mid m, n \geq 0, m \neq n\}.$$

A direct pumping argument is awkward because pumping may preserve the inequality. Closure properties reveal the problem immediately.

- Assume that $L_{\neq}$ is regular.
- The language

$$R = 0^* 1^*$$

is regular.

- Regular languages are closed under difference.
- Therefore, $R \setminus L_{\neq}$ must be regular.
- However,

$$R \setminus L_{\neq} = \{0^n 1^n \mid n \geq 0\},$$

which is not regular.

Contradiction. Therefore, $L_{\neq}$ is not regular.

Closure Proof 2: When Pumping Becomes Tedious

Consider

$$L_{<} = \{0^m 1^n \mid m < n\}.$$

Assume that $L_{<}$ is regular.

- Regular languages are closed under reversal.
- Swap 0 and 1 after reversing the strings.
- We would then obtain the regular language

$$L_{>} = \{0^m 1^n \mid m > n\}.$$

- Hence, the union

$$L_{<} \cup L_{>} = \{0^m 1^n \mid m \neq n\}$$

would be regular.

- Since $0^* 1^*$ is regular, its difference with this union would also be regular:

$$0^* 1^* \setminus (L_{<} \cup L_{>}) = \{0^n 1^n \mid n \geq 0\}.$$

This is impossible. Therefore, $L_{<}$ is not regular.

More Pumping-Lemma Examples I

Language	Choose s	Forced form of y	Pump and contradiction
$\{a^{n+1}b^n \mid n \geq 0\}$	$a^{p+1}b^p$	$y = a^k, k > 0$	Choose $i = 0$. Then $a^{p+1-k}b^p$ does not have exactly one more a than b .
$\{a^m b^n \mid m > n\}$	$a^{p+1}b^p$	$y = a^k, k > 0$	Choose $i = 0$. Then $p+1-k \leq p$, so the number of a s is no longer greater than the number of b s.
$\{a^n b^{2n} \mid n \geq 0\}$	$a^p b^{2p}$	$y = a^k, k > 0$	Choose $i = 0$. The counts become $p-k$ and $2p$, but $2p \neq 2(p-k)$.
$\{a^m b^n c^{m+n} \mid m, n \geq 0\}$	$a^p b^p c^{2p}$	$y = a^k, k > 0$	Choose $i = 0$. The first two blocks total $2p-k$, while the final block still has length $2p$.
$\{a^i b^j c^k \mid i = j \text{ or } j = k\}$	$a^p b^p c^{p+1}$	$y = a^r, r > 0$	Choose $i = 0$. Now $p-r \neq p$ and $p \neq p+1$, so neither equality holds.

More Pumping-Lemma Examples II

Language	Choose s	Forced form of y	Pump and contradiction
$\{u\#v \mid u = v \}$	$0^p\#1^p$	$y = 0^k, k > 0$	Choose $i = 0$. The two sides have lengths $p - k$ and p , so they are unequal.
$\{a^q \mid q \text{ is prime}\}$	a^q , where $q > p$ is prime	$y = a^k, 1 \leq k \leq p$	Choose $i = q + 1$. The new length is $q + qk = q(k + 1)$, which is composite.
$\{a^{2^n} \mid n \geq 0\}$	a^{2^m} , where $2^m > p$	$y = a^k, 1 \leq k \leq p$	Choose $i = 2$. Then $2^m < 2^m + k < 2^{m+1}$, so the new length is not a power of two.
$\{a^m b^n \mid m = n \text{ or } m = 2n\}$	$a^p b^p$	$y = a^k, k > 0$	Choose $i = 0$. Since $p - k < p$, the new number of a s is neither p nor $2p$.
$\{a^m b^{m+n} c^n \mid m, n \geq 0\}$	$a^p b^{2p} c^p$	$y = a^k, k > 0$	Choose $i = 0$. The two outer blocks total $2p - k$, but the middle block still has length $2p$.

Problems to Try at Home

For each language below, use the **five-step Pumping Lemma template** to prove that it is not regular.

- $L_1 = \{0^m 1^n \mid m \neq n\}$.
- $L_2 = \{0^m 1^n \mid \gcd(m, n) = 1\}$.
- $L_3 = \{0^m 1^n \mid m \mid n, m, n \geq 1\}$.
- $L_4 = \{(ab)^n c^n \mid n \geq 0\}$.
- $L_5 = \{a^{n!} \mid n \geq 1\}$.
- $L_6 = \{a^{F_n} \mid n \geq 0\}$, where F_n is the n th Fibonacci number.
- Let L_7 be the language of all correctly balanced parenthesis strings over $\{(,)\}$.

Summary

- **Intuition:** Finite states and sufficiently long inputs force cycles.
- **Theorem:** Every sufficiently long string in a regular language must be pumpable.
- **Proof strategy:** Choose a string that traps the pumpable part in a vulnerable region.
- **Rigor:** The argument must work for every valid decomposition $s = xyz$.
- **Alternative:** When direct pumping becomes awkward, closure properties may be cleaner.

Next Lecture: Context-Free Grammars and the idea of recursive structure.